In [2]:
```
# Single Inheritance- When a child class inherits only a single parent class.
# Example Code:
class Parent:
    def f1(self):
        print("This is function 1")
class Child(Parent):
    def f2(self):
        print("This is function 2")

obj1 = Child()
obj1.f2()
obj1.f1()
```

```
This is function 2
This is function 1
```

In [3]:
```
# Multiple Inheritance- When a child class inherits from more than one parent
# Example Code:
class Parent1:
    def f1(self):
        print("This is function 1")
class Parent2:
     def f2(self):
        print("This is function 2")
class Child(Parent1, Parent2):
    def f3(self):
        print("This is function 3")

obj1 = Child()
obj1.f1()
obj1.f2()
obj1.f3()
```

```
This is function 1
This is function 2
This is function 3
```

In [22]:
```
# Multilevel Inheritance : When a child class becomes a parent class for anoth
# Example Code:
class Parent:
    def __init__(self,__name):
        self.__name = __name

    def f1(self):
        print("This is Parent :",self.__name)

class Child1(Parent):
    def f2(self):
        print("This is Child1 of",self._Parent__name)

class Child2(Child1):
    def f3(self):
        print("This Child2 of",self._Parent__name) # accessing private variabl

obj1 = Child2("John")
obj1.f1()
obj1.f2()
obj1.f3()
```

```
This is Parent : John
This is Child1 of John
This Child2 of John
```

In [23]:
```
# Hierarchical Inheritance - involves multiple inheritance from the same base
 # Example Code:
class Parent:
    def __init__(self,name,age):
        self.name = name
        self.age = age

    def display(self):
        print("Name :",self.name,"Age :",self.age)

class Child1(Parent):
    def displayChild(self):
        super().display()
        print("Child1 of",self.name)

class Child2(Parent):
    def displayChild2(self):
        print("Child2's Parent's age is",self.age)

obj1 = Child1("Mary",20)
obj2 = Child2("Mary",20)

obj1.displayChild()
obj2.displayChild2()
```

```
Name : Mary Age : 20
Child1 of Mary
Child2's Parent's age is 20
```